MUC02 — Expected Outcomes

Credit Pre-Screening Engine — what a strong submission found

Use this deck to check your own notebook step by step. Your numbers will differ slightly depending on the cleaning choices you made — within about 2–3 percentage points is expected. What matters most is whether you reached the same conclusions, and whether you said what they mean for the bank.

Three things in the data needed a decision

441 missing incomes

Mostly Self-Employed and Business Owner applicants (11–13% of each). Fill with the median for that employment type — and note that missing income is itself a self-employed pattern.

1,039 blank credit scores

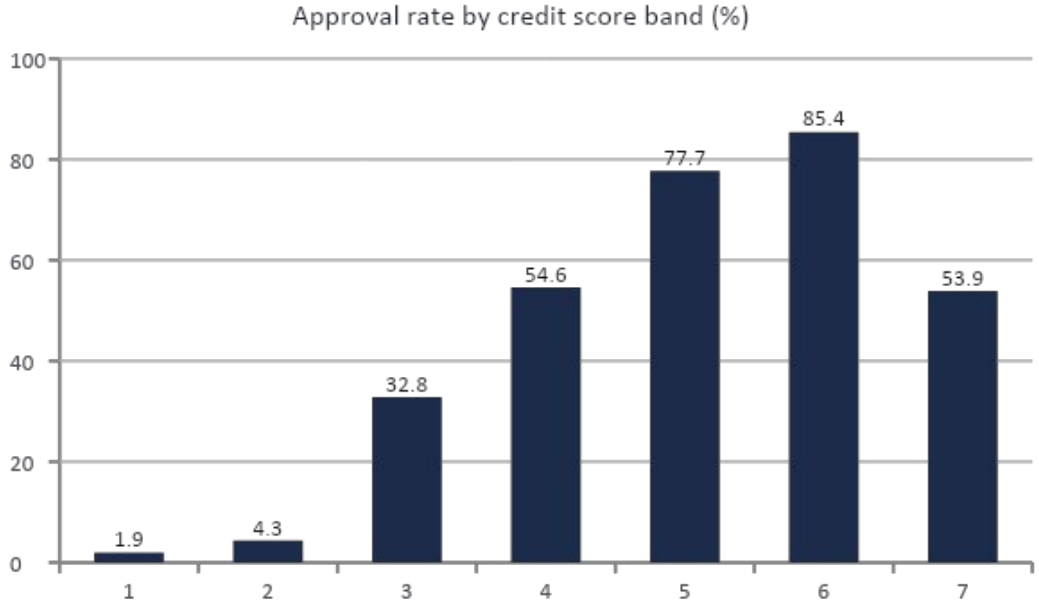
Not errors. These are new-to-credit applicants — credit_history_months is 0 for every one. Keep them and flag them with is_ntc. Dropping them removes 9% of applicants.

12 incomes above ₹6 lakh a month

All Salaried. Almost certainly annual income entered as monthly. Divide by 12 — or drop them, as long as you say so.

A cleaning log a bank reviewer could agree with is worth as much as the code.

Below a credit score of 650, the bank almost never approves



What a strong answer says

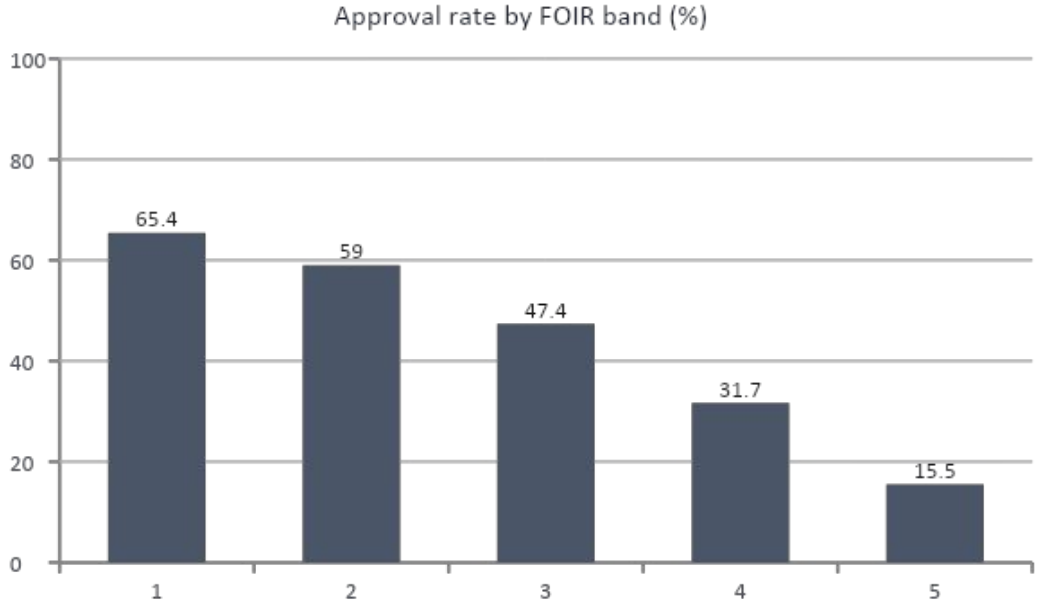
Approval jumps from 4.3% to 32.8% across the 650 line. The bank has an unwritten score floor — a rule a pre-screener can apply with confidence.

New-to-credit applicants are approved 53.9% of the time: the credit team is using judgement here.

Gig / Contract workers are approved least (34.9% vs 58.4% for Salaried).

"Lower scores get fewer approvals" describes the chart. "Below 650 the bank almost never approves" is a policy finding.

Once EMIs take more than half of income, approvals collapse



What a strong answer says

FOIR = (existing EMI + proposed EMI) ÷ monthly income.

Approval falls from 65.4% below 30% FOIR to 15.5% above 55%. The sharpest drop is past 50%.

A policy limit of 50–55% FOIR is what the historical decisions support.

FOIR turns out to be one of the three strongest drivers in the whole dataset — and you built it yourself.

79% accurate — but accuracy is not the number the bank cares about

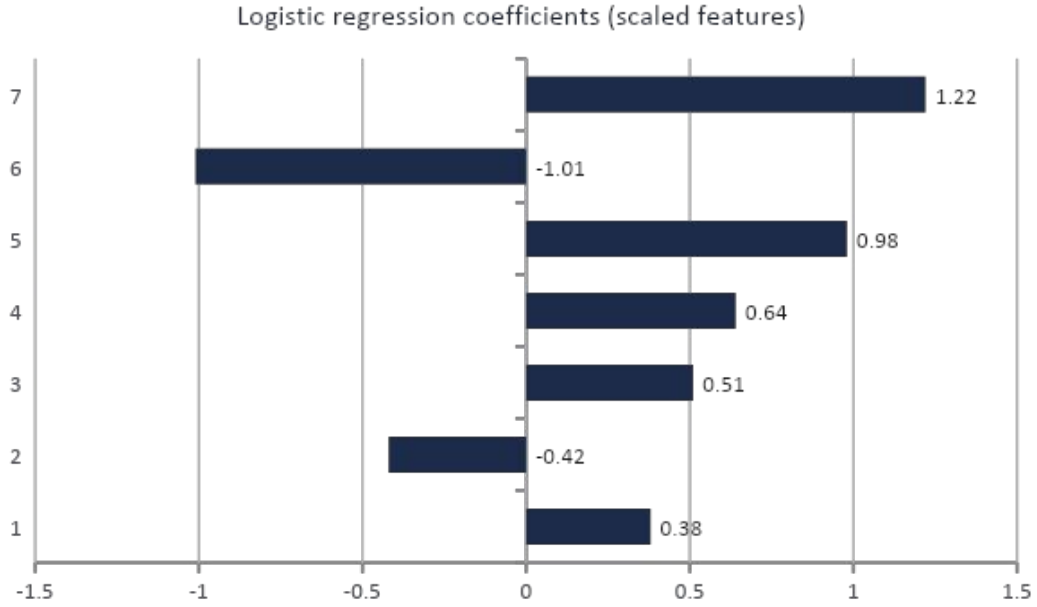
	Predicted: Approved	Predicted: Declined
Actually Approved	**1046** Correctly approved	**225** Good customers wrongly declined
Actually Declined	**277** Declines missed	**852** Correctly declined

Test set: 2400 applications · threshold 0.5

Metric	Value
Accuracy	79.1%
ROC-AUC	0.867
Precision (Declined)	79.1%
Recall (Declined)	75.5%
Accuracy — scored applicants	81.2%
Accuracy — new to credit	56.7%

At the default 0.5 cut-off, 225 applicants the bank approved would be turned away — and the model is barely better than a coin toss on new-to-credit applicants.

The model learned the same policy you found in Steps 5.2 and 5.3



Reading it like a credit analyst

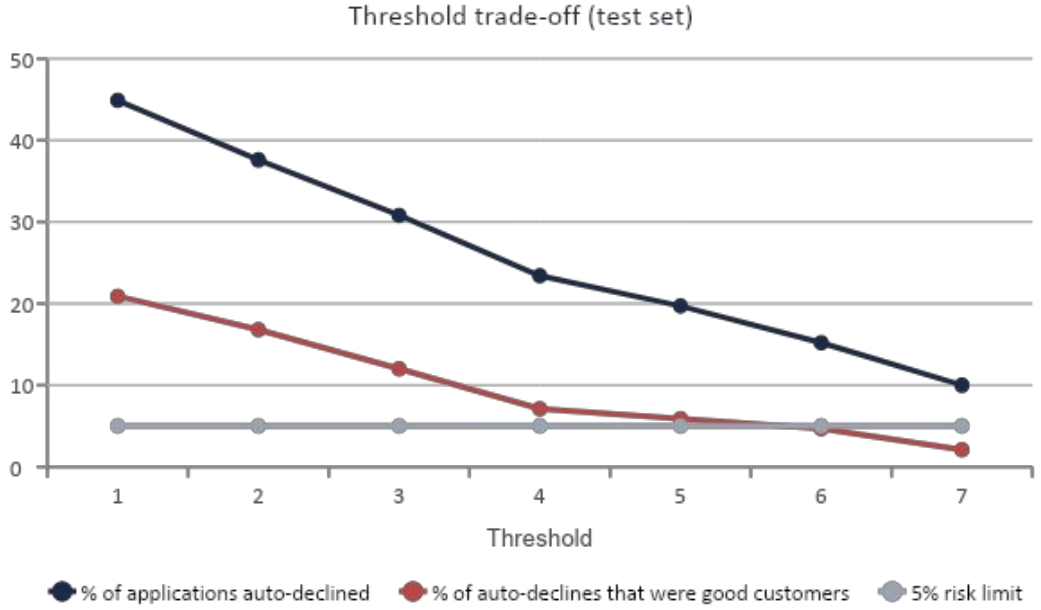
Positive pushes towards decline; negative towards approval.

Higher FOIR, more late payments and being a gig worker push towards decline. A higher credit score and a salaried job pull towards approval.

Because features were scaled, sizes can be compared directly.

"Every additional missed payment pushes an application towards decline" — say what the coefficient means for an applicant.

The bank sets the risk limit — the model shows where it falls



Threshold	Auto-declined	Wrong
0.50	44.9%	20.9%
0.60	37.6%	16.8%
0.70	30.8%	12%
0.80	23.4%	7.1%
0.85	19.7%	5.9%
0.90	15.2%	4.7%
0.95	10%	2.1%

0.90 is the lowest threshold within the 5% limit (4.7% wrong). At 0.5, one in five auto-declines would be a good customer.

What a 0.90 threshold is worth every month

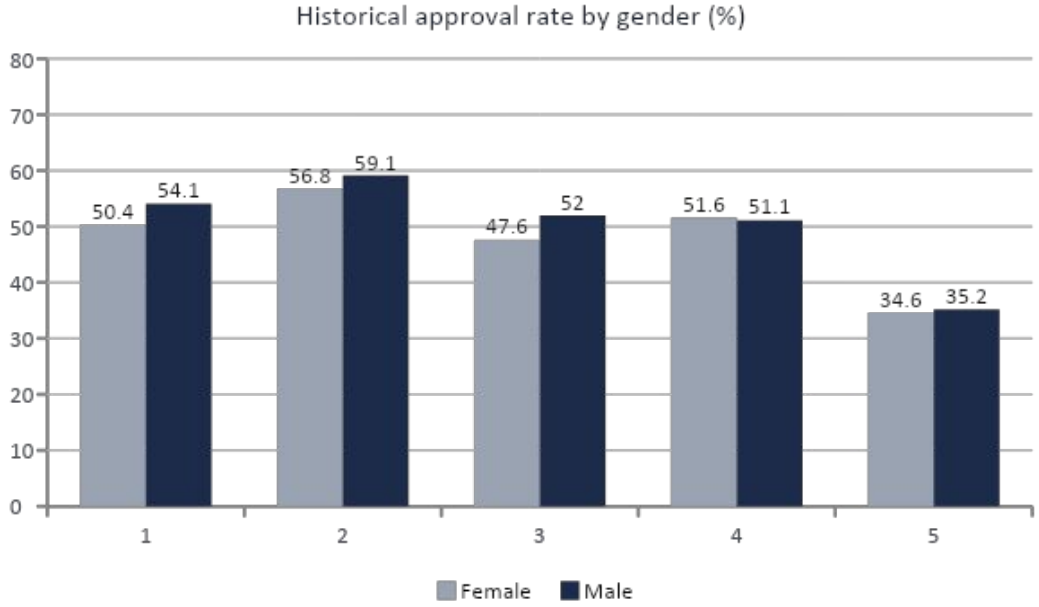
15.2%	~456	~190 hrs	4.7%
of applications auto-declined	applications a month (of 3,000)	analyst time freed (25 min each)	of auto-declines were good customers

Why the number is modest — and why that is fine

A single auto-decline lane at a strict risk limit catches only about 30.7% of all declines. That is still roughly 190 analyst hours a month with very little risk — a safe first step the bank can build on.

A strong briefing states the threshold, the applications and the hours together — in one sentence.

The raw gap is real — but it comes mostly from income and job mix



What a strong answer says

Overall: 54.1% male vs 50.4% female approved. Within Gig and Business Owner, the gap is under 1 point.

Female applicants have lower median income (₹45K vs ₹55K) and are almost twice as likely to be gig workers (21% vs 12%).

The model's auto-decline rate is almost identical by gender.

Leaving gender out is necessary, not sufficient. Income and job type can act as stand-ins — monitor outcomes by gender after launch.

What a distinction-level briefing sounds like

"I recommend auto-declining an application only when the model is at least 90% confident it will be declined. On our test data, that removes about 15.2% of applications — roughly 456 a month, or 190 analyst hours — and fewer than 5% of those would have been approved by the credit team. Two factors drive decisions most: FOIR, where approvals fall from 65.4% below 30% to 15.5% above 55%, and credit score, where approvals are under 5% below 650. The model should not decide on new-to-credit applicants — it is right only 56.7% of the time for them — so they should always go to an analyst. Gender is not used and auto-decline rates are almost identical by gender, but income and job type can act as proxies, so I would report outcomes by gender every month. Before going live, I would want repayment data: today the model copies past analyst decisions, not whether customers actually repaid."

Threshold · applications and hours · two drivers with numbers · a group to keep with humans · fairness position · one more piece of data.

The most common gaps — and what to do instead

What we often saw	What to do instead
Blank credit scores dropped	Keep new-to-credit applicants and flag them
Accuracy as the headline	State how many good customers are wrongly declined
Threshold left at 0.5	Use the table and the bank's 5% rule
Coefficients listed, not explained	Say what each one means for an applicant
"We did not use gender, so it is fair"	Measure outcomes by gender and think about proxies

What to carry into your next use case

1

A model's output is a probability. The decision is a threshold — and the business sets it.

2

Translate every result into the stakeholder's units: applications, hours, customers, rupees.

3

Always ask who the model should not decide for — and say so plainly.